Quantum Multi-Excitation Projective Simulation

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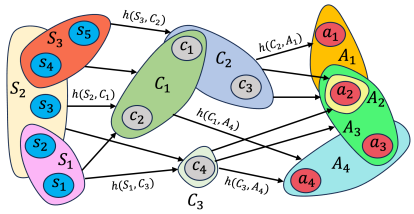
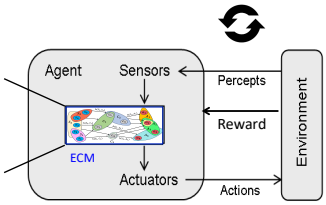
Basic Idea: Defining the multi-excitation projective simulation model in the quantum regime in the language of Fermion second-quantization. Explainable aspects of the model are highlighted and the analogy with quantum simulators is made, where a reinforcement learning task based on quantum simulation is presented.

Classical Multi-Excitation Projective Simulation (MEPS)

MEPS is an explainable, physically motivated learning model based on episodic and compositional memory (ECM) [1]. MEPS agents learn by first observing and then acting on their environment such that they receive a reward. Agents adapt to maximize their long-term expected reward.





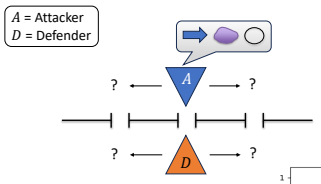
The ECM is represented as a clip network and decision-making is modeled as a random walk on the ECM.

Learning happens through updating ECM weights:

$$h^{(t+1)}(C_V, C_W) = 1 + (1-\gamma)\left(h^{(t)}(C_V, C_W) - 1\right) + R^{(t)} g^{(t)}(C_V, C_W)$$

where $g^{(t)}(C_V, C_W)$ informs the agent of which previous transitions led to rewards.

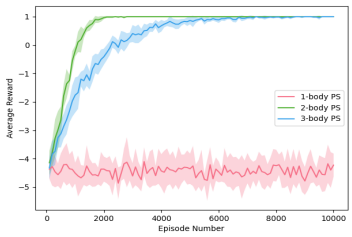
Invasion Game with Distraction



Task

D must guess which door A will go to and block them to get a reward.

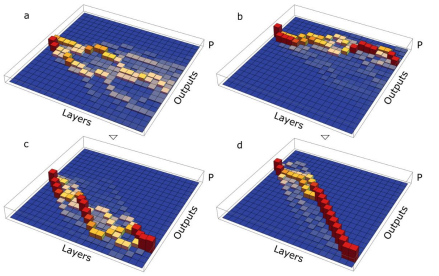
Similar behaviour expected from the quantum model for 2 layers.



Explainability

Quantum systems are notoriously delicate and opaque, which is in tension with explainability, so how can we recover some explainability in this regime?

Partial Traceability



Taken from [3] with permission

We can recover some notion of a "classical path" through partial traceability. In essence, it is a plot of the probability of observing excitations in each of the output modes, given a fixed input mode, for each layer of the ECM. Plots a-b) are before training while plots c-d) are after. Since each mode represents a clip, which themselves represent concepts, this method lets us understand the agent's chain-of-thought.

Modularity

Individual clip occupation number measurements are very destructive, so to avoid erasing most of the quantum information contained in the agent's ECM, we can instead measure the occupation number of collections of clips which we call modules. One can either prespecify these modules or allow them to arise through agent discovery, and they can be interpreted as sub-processes or tools that the agent uses in deliberation.

Dynamic Hypergraph

Dynamic hypergraphs are essentially a collection of ECMs each at some fixed time, so that the full set plays back the dynamic evolution of the agent's memory over the training period. If one were to record the h-values from the Hamiltonian, which is possibly less destructive and also more reproducible since the time-evolution operator doesn't change after training, then they could stitch them together and show how deliberation paths evolve and emerge over time.

Quantum MEPS

The Basics

Groups of clips are now represented by Fermionic many-body Fock states $|C_i\rangle = \hat{C}_i^\dagger |vac\rangle$ with creation operators $\hat{C}_i^\dagger \equiv \hat{c}_{i_1}^\dagger \cdots \hat{c}_{i_N}^\dagger$ indicating the "activation" of clip groups during deliberation.

The connectivity of the agent's ECM is encoded in the second-quantized Hamiltonian

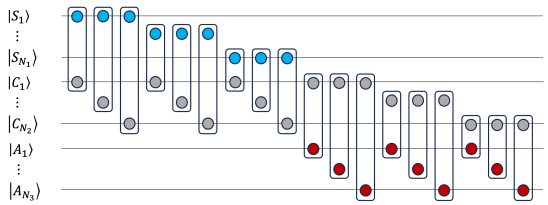
$$\hat{H} = \sum_{l,o} \sum_{j_1,\ldots,j_l} \sum_{k_1,\ldots,k_o} h_{j_1,\ldots,j_l,k_1,\ldots,k_o} \hat{C}_{k_1}^\dagger \cdots \hat{C}_{k_o}^\dagger \hat{C}_{j_1} \cdots \hat{C}_{j_l}$$

where $h_{j_1,\ldots,j_l,k_1,\ldots,k_o}$ are the trainable parameters that correspond to the coupling strengths in the agent's ECM.

Model Architecture

$\{S_i\}$ = percept hyperedges
$\{C_i\}$ = intermediate clip hyperedges
$\{A_i\}$ = action hyperedges

$$\hat{U}_{model}(t) = \prod_k \hat{U}_k(\Delta t_k)$$



Learning

The classical MEPS update rule relies on having well-defined ECM trajectories but we have superpositions of trajectories in the quantum model, so how does the agent learn?

→ Variationally, by minimizing a loss function which encodes the classical MEPS update rule.

Deliberation

The classical random walk through the ECM representing deliberation is now replaced with a quantum walk of multiple excitations time-evolving through a sequence of intervals Δt_k according to each $\hat{U}_k(\Delta t_k)$, connecting percepts to intermediate clips to actions.

↳ There is potential for some excitations not to make it to the action modes, so one idea is to repeatedly measure the total particle number corresponding to the designated output modes of the next layer for a given $\hat{U}_k$ until the occupation number matches the number of initial excitations. The high degeneracy of this measurement will preserve much of the entanglement. Then repeat for every layer.

Quantum Simulation as Reinforcement Learning

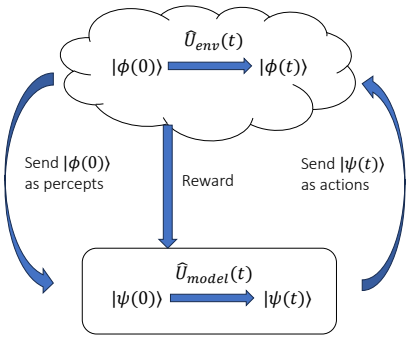
Second-quantized Fermionic quantum MEPS model is essentially a quantum simulator, so simulation is a natural learning task.

Task

Emulate unknown $\hat{U}_{env}(t)$ with $\hat{U}_{model}(t)$ as closely as possible given only $|\phi(0)\rangle$ and a reward based on fidelity between $|\phi(t)\rangle$ and $|\psi(t)\rangle$.

- An example of an agent and environment that are both quantum, who can also exchange quantum data.
- Classical pre- and post-processing can be added for better interpretability.

Potential to learn quantum circuits and gates in terms of many-body time-evolution operators.



References

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3. F. Flamini et al., arXiv preprint arXiv: 2301.13669 (2023).
4. I. M. Georgescu et al., Rev. Mod. Phys. 86, 153-185 (2014).

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